IP DAY 2026

DETAILED GUIDELINES

ROBO RACE

FIELD SPECIFICATION

- Closed loop track, approx. 40 m per lap (the race may involve multiple laps)
- Uneven surface (soil/sand/grits) + Obstacles
- Multiple Curves Ramp/Slope Sections
- Note: Teams are advised to design their vehicles and control systems accordingly.

TIME PENALTIES

- Result / Penalty
- Touch/Cross Boundary Line
- Restart from last Checkpoint (Clock keeps running)
- Touching Static Obstacle - +5 Seconds per touch
- Skipping a Checkpoint - +30 Seconds
- Skipping the Ramp/Sand - +30 Seconds

CONTACT

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https://maps.app.goo.gl/VEQJTP4tNcQue2bY8t

TASKS TO DO

- Closed loop track, approx. 40 m per lap (the race may involve multiple laps)
- Uneven surface (soil/sand/grits) + Obstacles
- Multiple Curves Ramp/Slope Sections
- Note: Teams are advised to design their vehicles and control systems accordingly.
- Line Faults: If any part of the robot touches the boundary line or goes completely out of the track, it is a foul.
- The Restart Rule: Upon a foul, the robot must be picked up and placed back at the last passed checkpoint.
- Timer Status: The master clock does not stop during a restart. The time taken to reset the robot is effectively your penalty.
- The Incline (Ramp): A 30°–45° slope. The robot must climb and descend without falling off the sides.
- The Pit (Sand & Soil): A section filled with loose soil/sand to test traction and motor torque.
- The Maze (Avoidance): Static blocks or pillars. Touching an obstacle results in a 5-second penalty added to the lap time.
- Task/Checkpoint Skip: If a team chooses to skip a difficult obstacle (like the ramp) or a checkpoint, a 30-second penalty is added to the total lap time.
- Maximum Skips: A robot is allowed only one skip per lap. Skipping more than one checkpoint results in a DNF (Did Not Finish) for that lap.

ROBOT SPECIFICATIONS

- RC Car Configuration & Specifications Parameter Allowed Range Vehicle Scale Approx. 1:16 or 1:18.
- Ground Clearance (likely as per your requirement)
- Motor 20,000 RPM (Brushed) Maximum

  OR
- 3000 kV (Brushless) Maximum
- ESC 25 A maximum or Motor driver (likely as per your requirement)
- Battery 8V to 9V max (2S only)

IMPORTANT NOTES FOR PARTICIPANTS

- Participants may bring charger, extra battery as required.
- Organiser will NOT be responsible for any damage to the robot.
- Only electric RC vehicles are allowed; nitro or any fuel-based RC cars are NOT permitted.
- Commercially available RC cars are "NOT" allowed, And any modifications beyond the specified limits are strictly prohibited.